

# Xianzhe Wang

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## EDUCATION

**University of California, Davis** *PhD Student in Transportation Technology and Policy* 09/2025-Present  
GPA: 4.0/4.0.    **Advisor:** Prof. Yueyue Fan  
**Research Focus:** Decision Aware Origin Destination Demand Estimation with Physical Consistency Constraints  
**Beijing University of Chemical Technology** *MS in Management Science and Engineering* 09/2022-06/2025  
GPA: 3.5/4.0.    **Advisor:** Prof. Xiang Li  
**Thesis:** Dynamic Repositioning Optimization for Bike-Sharing Systems with Spatiotemporal Demand Forecasting  
**East China University of Science and Technology** *BA in Logistics Management* 09/2017-06/2021  
GPA: 3.1/4.0.    **Advisor:** Dr. Jun Chen  
**Thesis:** Drone-Assisted Reverse Logistics Route Planning for Fresh Produce under Uncertainty

## RESEARCH EXPERTISE

- **Methodologies:** Stochastic Optimization, Decomposition Algorithms, Sequential Decision-Making, Decision Aware Estimation, and Learning Augmented Optimization.
- **Applications:** Supply Chain and Transportation Systems, with emphasis on Network Design, Demand and Capacity Planning, Routing and Scheduling, and Resource Allocation.

## PUBLICATIONS

- **Xianzhe Wang**, Yueyue Fan\*, and Johannes O. Royset. "Decision-Aware Origin–Destination Demand Estimation with Structurally Constrained Data." *Under review at Transportation Science*.
- Spyros Garyfallos, Yunan Liu\*, **Xianzhe Wang**, Pere Barlet-Ros, and Alberto Cabellos-Aparicio. "Solving Nonstationary Non-Markovian Queues: A Transfer Learning Neural Network Approach." *Under review at INFORMS Journal on Computing*.
- Xiang Li, **Xianzhe Wang**, and Ziyang Feng\*. "Dynamic repositioning in bike-sharing systems with uncertain demand: An improved rolling horizon framework." *Omega*, vol. 126, 2024, 103047.
- Xiang Li, Nan Chen, Hongguang Ma\*, Fapeng Nie, and **Xianzhe Wang**. "A parallel genetic algorithm with variable neighborhood search for the vehicle routing problem in forest fire-fighting." *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 10, 2024, pp. 14359–14375.
- Ziyang Feng, **Xianzhe Wang**, and Xiang Li\*. "AI-Driven Subway Passenger Demand Prediction: Algorithm Innovation and Practice." *Frontiers of Science and Technology of Engineering Management*, 44(6): 32–39, 2025. (in Chinese with English abstract) doi:10.11847/fj.44.6.32.

## PROFESSIONAL EXPERIENCE

- Teaching Assistant** (*University of California, Davis*) **Davis** 09/2026 - 12/2026
- Led lab and discussion sections for ECI 153 (**Deterministic Optimization & Design**), covering linear, nonlinear, and dynamic programming.
  - Guided students in optimization modeling, computational implementation, and engineering applications.
- Research Internship** (*SRIBD & The Chinese University of Hong Kong, Shenzhen*) **Shenzhen** 09/2024 - 05/2025
- Integrated **Learning to Optimize** techniques into the branch-and-bound framework for mixed-integer optimization.
  - Investigated **performance variability** to improve solver efficiency, particularly during presolve.

**Research Assistant** (*Institute for AI Industry Research, Tsinghua University*) **Beijing** 12/2023 - 05/2024

- Developed a maritime rescue scheduling system integrating vessel dispatch and adaptive search-pattern decisions.
- Designed a **migration-based evolutionary algorithm** for efficient rescue scheduling and resource allocation.

**Associate Editor** (*Xiaoyao Technology, Top 10 AI Technology Media in China*) **Beijing** 03/2021 - 08/2025

- Authored and edited nearly **100 technical articles** on NLP, LLMs, AI Agents, and related AI topics.
- Reached **44,000+** readers for a single article and **200,000+ total readers**, with selected technical articles reprinted and shared by research labs at **Fudan University, Renmin University of China**, and other academic institutions.

**AI Research Camp Member** (*DeeCamp 2021, Organized by Kai-Fu Lee*) **Beijing** 06/2021 - 08/2021

- Developed an **AI-based document extraction pipeline** using image segmentation, OCR, and language models.
- Converted bond prospectuses into structured data, extracting **14 key fields** with **94% accuracy**.

## SELECTED RESEARCH PROJECTS

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**California Air Resources Board Incentive Program Impact Assessment** 2026–Present  
*University of California, Davis* PI: Yueyue Fan

- Conducted county level impact analysis of California clean air incentive programs, integrating emissions, air quality, enforcement, wildfire, socioeconomic, and land use data to evaluate program effectiveness and regional disparities.

**Theory and Methods for Operating and Managing Shared Transportation Modes** 2022–2024  
*Beijing University of Chemical Technology* PI: Xiang Li

- Conducted research on bike-sharing systems, including spatiotemporal demand forecasting, dynamic repositioning modeling, and algorithm design, and translated model insights into operational and policy recommendations for shared mobility management.

**Big Data-Driven Collaborative Decision-Making for Urban Transportation** 2022–2024  
*Beijing University of Chemical Technology* PI: Xiang Li

- Conducted research on data-driven algorithms for large-scale urban transportation systems, with emphasis on parallel computing, reinforcement learning, and collaborative decision-making for efficient transportation operations.

## CONFERENCE PRESENTATIONS

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**Accepted for Presentation** (*2026 INFORMS Annual Meeting*) **San Francisco, United States** 11/2026

- **Presentation:** “Decision-Aware Origin-Destination Demand Estimation with Structurally Constrained Data.”

**Accepted for Presentation** (*Third International Conference on Clean Mobility and Energy*) **Ghent, Belgium** 09/2026

- **Presentation:** “Evaluating the Effectiveness of Clean Air Subsidies: Evidence on Outcomes, Enforcement, and Regional Spillovers from California’s Carl Moyer Program”

**Presented** (*5th Conference on Intelligent Decision-Making and Games*) **Guilin, China** 05/2024

- **Presentation:** “Dynamic Repositioning in Bike-Sharing Systems with Uncertain Demand: An Improved Rolling-Horizon Framework.”

## SKILLS & TOOLS

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|----------------------------------------------------|---------------------------------------------------------|
| • <b>Programming:</b> Python, Matlab, R, C/C++     | • <b>ML &amp; LLM:</b> PyTorch, Transformers, LangChain |
| • <b>Optimization Solvers:</b> Gurobi, CPLEX, COPT | • <b>Development Tools:</b> Git, Linux, LaTeX           |

## SELECTED HONORS AND AWARDS

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- **National Scholarship** (*Top Scholarship in China; 0.2% domestically*)
- **Academic Scholarship**, Beijing University of Chemical Technology(*First Prize*)
- **Excellent Undergraduate Thesis**, East China University of Science and Technology(*Top 5%*)
- **Comprehensive Course Scholarship**, East China University of Science and Technology(*Third Prize*)
- **7th Young Leaders Action Learning Program**, East China University of Science and Technology(*First Prize*)